

Differentiation of Exponential Functions and Applications

1. Find $\frac{dy}{dx}$ for each of the following:

(a) $y = e^x - 3$

(b) $y = 5e^x + 2x$

(c) $y = (e^x + 7)^3$

(d) $y = xe^x$

(e) $y = \frac{e^x}{x}$

(f) $y = \sqrt{2e^x - 5}$

(a) $\frac{dy}{dx} = e^x$ (b) $\frac{dy}{dx} = 5e^x + 2$ (c) $\frac{dy}{dx} = 3(e^x + 7)^2 \cdot e^x$

(d) $\frac{dy}{dx} = x \cdot e^x + e^{x-1} = e^x(x+1)$ (e) $y = \frac{e^x}{x}$
 $\frac{dy}{dx} = \frac{x e^x - e^x}{x^2} = \frac{e^x(x-1)}{x^2}$ (f) $y = (2e^x - 5)^{\frac{1}{2}}$
 $\frac{dy}{dx} = \frac{1}{2} (2e^x - 5)^{-\frac{1}{2}} \cdot (2e^x)$
 $\frac{dy}{dx} = \frac{e^x}{\sqrt{2e^x - 5}}$

2. If $y = e^{kx}$ is a solution of the equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$ find the possible values of the constant k .

$y = e^{kx}$
 $\frac{dy}{dx} = k \cdot e^{kx}$
 $\frac{d^2y}{dx^2} = k^2 \cdot e^{kx}$

$k^2 \cdot e^{kx} + k \cdot e^{kx} - 6e^{kx} = 0$
 $e^{kx} (k^2 + k - 6) = 0$
 $\neq 0 \quad (k+3)(k-2) = 0$
 $k = -3 \text{ or } 2$

Differentiation of Exponential Functions and Applications continued

3. Find the equation of the tangent and the normal to the curve $y = 2e^{3x} - 1$ at the point where $x = 0$.

$$x=0, y = 2e^{3(0)} - 1 = 1 \rightarrow (0, 1)$$

$$y = 2e^{3x} - 1$$

$$y' = 2e^{3x} \cdot 3 = 6e^{3x}$$

$$y' = 6e^{3x}$$

$$x=0, m_T = 6e^{3(0)} = 6$$

$$m_N = -\frac{1}{6}$$

Tangent: $y - 1 = 6(x - 0)$
 $y = 6x + 1$ ✓

Normal: $y - 1 = -\frac{1}{6}(x - 0)$
 $y = -\frac{x}{6} + 1$ ✓

4. The price \$p\$ of a lipstick is given by $p = 10e^{-x}$ for $0 < x < 2$ where x is the number of lipsticks (in thousands) that can be sold in a week.

- (a) Revenue is defined as the total dollar amount received for the sale of x lipsticks. Find the price at which the weekly revenue will be a maximum.
(b) Find the maximum weekly revenue. x is measured in thousands

(a) Revenue: $R(x) = (1000x)(10e^{-x}) = 10000xe^{-x}, 0 < x < 2$

$$R'(x) = 10000x \cdot e^{-x}(-1) + 10,000e^{-x}$$

$$R'(x) = 10,000e^{-x}(1-x)$$

$$R'(x) = 0 \Rightarrow 10,000e^{-x}(1-x) = 0$$

$$1-x = 0$$

$$x = 1$$

Nature:

$$R'(x) = 10,000e^{-x}(1-x)$$

x	$\frac{1}{2}$	1	1.5
$R'(x)$	+	0	-

Maximum Revenue at $x = 1$

$$\text{Price, } P = 10e^{-x} = 10e^{-1} = \frac{10}{e} \approx \$3.68$$

(b) Maximum Revenue, $R(x) = 10000xe^{-x}$
 $x=1, R(1) = 10000 \cdot 1 \cdot e^{-1} = \frac{10000}{e} \approx \$3678.79/\text{week}$

Differentiation of Exponential Functions and Applications continued

5. Consider the function $y = xe^{-x}$.

(a) Find the coordinates of the stationary point and determine its nature.

(b) Find the value of x for which:

(i) $y > 0$ (ii) $\frac{dy}{dx} > 0$ (iii) $\frac{d^2y}{dx^2} > 0$

(c) Given that $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$ write down the equation of the horizontal asymptote of the function $y = xe^{-x}$.

(d) Sketch the graph of the function $y = xe^{-x}$.

(c) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$
 $\therefore y = 0$ (x -axis) is the H.A.

(a) $y = xe^{-x}$

$y' = x \cdot e^{-x}(-1) + e^{-x} \cdot 1$

$y' = e^{-x}(1-x)$

S.P.: $y' = 0$

$e^{-x}(1-x) = 0$

$\neq 0 \quad x = 1, y = 1 \cdot e^{-1} = \frac{1}{e}$

$(1, \frac{1}{e})$ is the stationary point

relative: $y' = e^{-x}(1-x)$

$y'' = e^{-x}(-1) + (1-x)e^{-x}(-1)$

$= e^{-x}(-1-1+x)$

$y'' = e^{-x}(x-2)$

at $x=1, y'' = e^{-1}(1-2) = -\frac{1}{e} < 0$

Maximum T.P. at $(1, \frac{1}{e})$

(b) $y = xe^{-x}$

(i) $y > 0$

$\frac{x}{e^x} > 0$

$x > 0$

(ii) $\frac{dy}{dx} > 0$

$e^{-x}(1-x) > 0$

$1-x > 0$

$-x > -1$

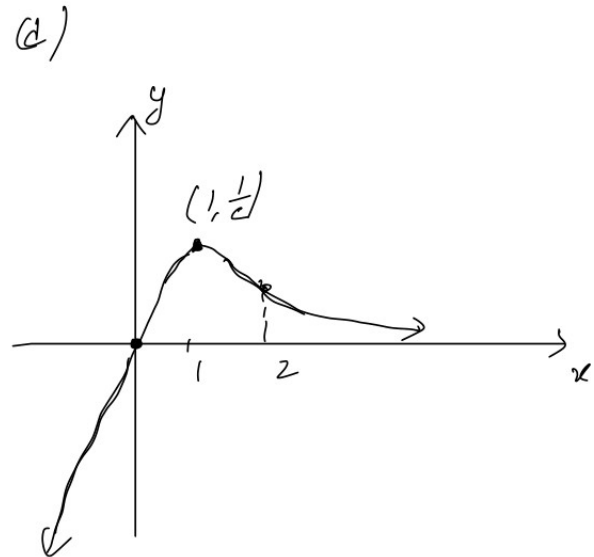
$x < 1$

(iii) $\frac{d^2y}{dx^2} > 0$

$e^{-x}(x-2) > 0$

$x-2 > 0$

$x > 2$



Differentiation of Logarithmic Functions

6. Find $\frac{dy}{dx}$:

(a) $y = \ln 3x$

(c) $y = \ln(x^2 - 3x + 2)$

(e) $y = x \ln x$

(b) $y = \ln(x^2 - 5)$

(d) $y = \ln(1 - x^3)$

(f) $y = e^x \ln x$

(a) $y = \ln 3x$
 $y = \ln 3 + \ln x$
 $y' = 0 + \frac{1}{x} = \frac{1}{x}$

(d) $y = \ln(1 - x^3)$
 $y' = \frac{-3x^2}{1 - x^3}$

(b) $y = \ln(x^2 - 5)$
 $y' = \frac{2x}{x^2 - 5}$

(c) $y = x \cdot \ln x$
 $y' = x \cdot \frac{1}{x} + \ln x \cdot 1 = 1 + \ln x$

(c) $y = \ln(x^2 - 3x + 2)$
 $y' = \frac{2x - 3}{x^2 - 3x + 2}$

(f) $y = e^x \cdot \ln x$
 $y' = e^x \cdot \frac{1}{x} + \ln x \cdot e^x$
 $y' = e^x \left(\frac{1}{x} + \ln x \right)$

7. Differentiate with respect to x :

(a) $\ln x^9$

(c) $\ln\left(\frac{x}{2}\right)$

(e) $\ln\sqrt{x+5}$

(b) $\ln(5x-7)^8$

(d) $\ln\left(\frac{7x-6}{x+8}\right)$

(f) $\ln\sqrt{x^2-1}$

(a) $y = \ln x^9$
 $y = 9 \ln x$
 $y' = \frac{9}{x}$

(d) $y = \ln\left(\frac{7x-6}{x+8}\right)$
 $y = \ln(7x-6) - \ln(x+8)$
 $y' = \frac{7}{7x-6} - \frac{1}{x+8}$

(b) $y = \ln(5x-7)^8$
 $y = 8 \ln(5x-7)$
 $y' = 8 \cdot \frac{5}{5x-7} = \frac{40}{5x-7}$

(e) $y = \ln\sqrt{x+5} = \ln(x+5)^{\frac{1}{2}} = \frac{1}{2} \ln(x+5)$
 $y' = \frac{1}{2} \cdot \frac{1}{x+5} = \frac{1}{2(x+5)}$

(c) $y = \ln\left(\frac{x}{2}\right)$
 $y = \ln x - \ln 2$
 $y' = \frac{1}{x} - 0$
 $y' = \frac{1}{x}$

(f) $y = \ln\sqrt{x^2-1}$
 $= \ln(x^2-1)^{\frac{1}{2}}$
 $y = \frac{1}{2} \ln(x^2-1)$
 $y' = \frac{1}{2} \cdot \frac{2x}{x^2-1} = \frac{x}{x^2-1}$

Differentiation of Logarithmic Functions continued

8. If $f(x) = (x+1)\ln x$ find $f'(1)$ and $f''(1)$.

$$f'(x) = (x+1) \cdot \frac{1}{x} + \ln x \cdot 1 = 1 + \frac{1}{x} + \ln x$$

$$f'(x) = 1 + \frac{1}{x} + \ln x, \quad x=1, \quad f'(1) = 1 + \frac{1}{1} + \ln(1) = 2$$

$$\therefore \underline{\underline{f'(1) = 2}}$$

$$f''(x) = 0 + \left(-\frac{1}{x^2}\right) + \frac{1}{x}$$

$$f''(x) = -\frac{1}{x^2} + \frac{1}{x}$$

$$x=1, \quad f''(1) = -\frac{1}{1^2} + \frac{1}{1} = -1 + 1 = 0, \quad \therefore \underline{\underline{f''(1) = 0}}$$

9. Use the Change of Base result to differentiate:

(a) $\log_7 x$

$$y = \log_7 x = \frac{\ln x}{\ln 7}$$

$$y' = \frac{1}{\ln 7} \cdot \frac{1}{x}$$

$$y' = \frac{1}{x \ln 7}$$

(b) $\log_{10}(3x-5)$

$$y = \log_{10}(3x-5)$$

$$y = \frac{\ln(3x-5)}{\ln 10}$$

$$y' = \frac{1}{\ln 10} \cdot \frac{3}{3x-5}$$

$$y' = \frac{3}{(3x-5) \ln 10}$$

(c) $\log_5(x^2+1)$

$$y = \log_5(x^2+1) = \frac{\ln(x^2+1)}{\ln 5}$$

$$y' = \frac{1}{\ln 5} \cdot \frac{2x}{x^2+1}$$

$$y' = \frac{2x}{(x^2+1) \ln 5}$$

10. (a) Find the derivative of a^x using the identity $a^x = e^{\ln a^x} = e^{x \ln a}$.

(b) Hence find the derivative of:

(i) 7^x

$$(i) \quad \underline{\underline{7^x \ln 7}}$$

(ii) 9^{2x-5}

$$= 9^{2x-5} \cdot \ln 9 \cdot 2$$

$$= \underline{\underline{2 \ln 9 \cdot 9^{2x-5}}}$$

Integration Involving Exponential Functions and Applications

11. Find the following indefinite integrals:

$$(d) \int 6xe^{x^2} dx$$

$$(e) \int xe^{x^2} dx$$

$$(f) \int 6x^2e^{x^3} dx$$

$$(g) \int (x+1)e^{x^2+2x} dx$$

$$(h) \int \frac{2}{\sqrt{x}e^{\sqrt{x}}} dx$$

$$(i) \int \sqrt{e^x} dx$$

$$(d) \int 6x \cdot e^{x^2} dx = 3 \int 2x \cdot e^{x^2} dx = 3 \cdot e^{x^2} + C$$

$$(e) \frac{1}{2} \int 2x \cdot e^{x^2} dx = \frac{1}{2} e^{x^2} + C$$

$$(f) \int 6x^2 \cdot e^{x^3} dx = 2 \int 3x^2 \cdot e^{x^3} dx = 2 \cdot e^{x^3} + C$$

$$(g) \frac{1}{2} \int 2(x+1) e^{x^2+2x} dx = \frac{1}{2} e^{x^2+2x} + C$$

$$(h) \int \frac{2}{\sqrt{x}e^{\sqrt{x}}} dx = -2 \int \frac{1}{2\sqrt{x}} \cdot e^{-\sqrt{x}} dx = -4 \cdot e^{-\sqrt{x}} + C = \frac{-4}{e^{\sqrt{x}}} + C$$

$$(i) \int \sqrt{e^x} dx = \int (e^x)^{\frac{1}{2}} dx = \int e^{\frac{1}{2}x} dx = 2 \cdot e^{\frac{1}{2}x} + C$$

Integration Involving Exponential Functions and Applications continued

12. Evaluate each of the following definite integrals:

$$(a) \int_1^2 3e^{4x} dx$$

$$(b) \int_0^1 4x^3 e^{2x^4} dx$$

$$(c) \int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$(d) \int_1^3 3e^{2x} (e^{-2x} - 1) dx$$

$$(e) \int_{-1}^1 \frac{dx}{e^{2-3x}}$$

$$(a) \int_1^2 3e^{4x} dx = 3 \left[\frac{e^{4x}}{4} \right]_1^2 = \frac{3}{4} [e^8 - e^4]$$

$$(b) \frac{1}{2} \int_0^1 2 \cdot 4x^3 \cdot e^{2x^4} dx = \frac{1}{2} [e^{2x^4}]_0^1 = \frac{1}{2} [e^2 - 1]$$

$$(c) 2 \int_1^4 \frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}} dx = 2 [e^{\sqrt{x}}]_1^4 = 2[e^2 - e]$$

$$(d) \int_1^3 3e^{2x} (e^{-2x} - 1) dx = \int_1^3 (3 - 3e^{2x}) dx = \left[3x - \frac{3e^{2x}}{2} \right]_1^3 = \left(9 - \frac{3e^6}{2} \right) - \left(3 - \frac{3e^2}{2} \right) \\ = 6 - \frac{3}{2}e^6 + \frac{3}{2}e^2 \\ = 6 - \frac{3}{2}(e^6 - e^2)$$

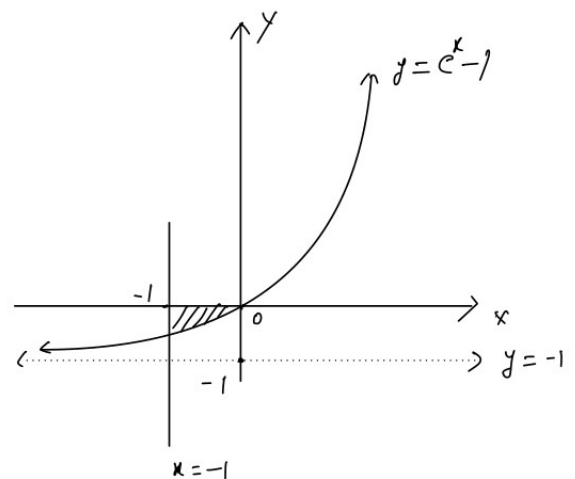
$$(e) \int_{-1}^1 \frac{dx}{e^{2-3x}} = \int_{-1}^1 (e^{2-3x})^{-1} dx \\ = \int_{-1}^1 e^{3x-2} dx = \left[\frac{e^{3x-2}}{3} \right]_{-1}^1 \\ = \frac{1}{3} [e - e^{-5}]$$

13. Sketch the graph of $y = e^x - 1$ and hence find the area bounded by the curve, the x -axis and the line $x = -1$.

$$\text{Area, } A = \left| \int_{-1}^0 (e^x - 1) dx \right| \\ = \left| [e^x - x]_{-1}^0 \right| \\ = \left| (1 - 0) - (e^{-1} + 1) \right| \\ = \left| 1 - \frac{1}{e} - 1 \right|$$

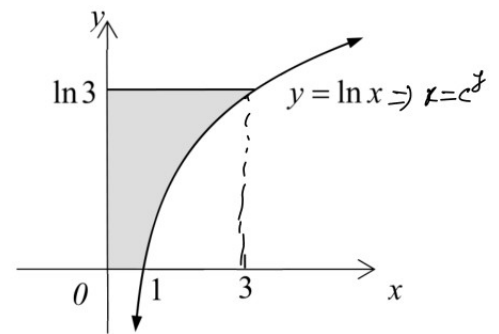
$$A = \left| -\frac{1}{e} \right|$$

$$A = \frac{1}{e} \text{ units}^2$$



Integration Involving Exponential Functions and Applications continued

14. The diagram on the right shows the region bounded by the curve $y = \ln x$, the coordinate axes and the line $y = \ln 3$.



- (a) Find the area of the shaded region.
- (b) Hence find the exact value of $\int_1^3 \ln x \, dx$.
- (c) Find the exact volume of the solid formed by rotating the shaded region about the y-axis.

(a) Area, $A = \int_0^{\ln 3} e^y \, dy$

$$= [e^y]_0^{\ln 3}$$

$$= e^{\ln 3} - e^0$$

$$= 3 - 1$$

$$A = 2 \text{ units}^2$$

(b) $\int_1^3 \ln x \, dx = 3 \cdot \ln 3 - 2$

(c) Volume, $V_y = \pi \int_0^{\ln 3} (e^y)^2 \, dy$

$$= \pi \int_0^{\ln 3} e^{2y} \, dy$$

$$= \pi \left[\frac{e^{2y}}{2} \right]_0^{\ln 3}$$

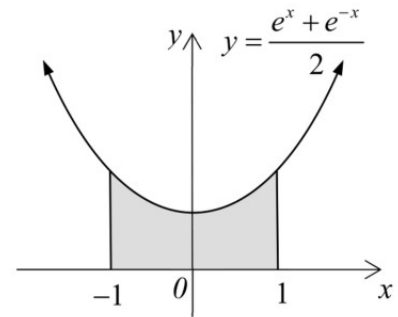
$$= \frac{\pi}{2} [e^{2 \ln 3} - e^{2(0)}]$$

$$= \frac{\pi}{2} [9 - 1]$$

$$V = 4\pi \text{ units}^3$$

Integration Involving Exponential Functions and Applications continued

15. (a) Find the area of the region bounded by the curve $y = \frac{e^x + e^{-x}}{2}$, the lines $x = -1$ and $x = 1$ and the x -axis as shown in the diagram on the right.



- (b) Find the volume of the solid of revolution obtained when this region is rotated about the x -axis.

$$\textcircled{a} \quad \text{Area, } A = \int_{-1}^1 \left(\frac{e^x + e^{-x}}{2} \right) dx \quad \xrightarrow{\text{symmetry}} \quad 2 \cdot \int_0^1 \left(\frac{e^x + e^{-x}}{2} \right) dx$$

$$= \left[e^x + \frac{e^{-x}}{-1} \right]_0^1 = \left[e^x - e^{-x} \right]_0^1 = (e - e^{-1}) - (1 - 1)$$

$$A = \left(e - \frac{1}{e} \right) \text{ units}^2$$

$$\textcircled{b} \quad \text{Volume, } V_x = \pi \int_{-1}^1 \left(\frac{e^x + e^{-x}}{2} \right)^2 dx \quad \xrightarrow{\text{symmetry}} \quad 2\pi \int_0^1 \left(\frac{e^{2x} + 2 + e^{-2x}}{4} \right) dx$$

$$V_x = \frac{\pi}{2} \left[\frac{e^{2x}}{2} + 2x + \frac{e^{-2x}}{-2} \right]_0^1$$

$$= \frac{\pi}{4} \left[e^{2x} + 4x - e^{-2x} \right]_0^1$$

$$= \frac{\pi}{4} \left[(e^2 + 4 - e^{-2}) - (1 + 0 - 1) \right]$$

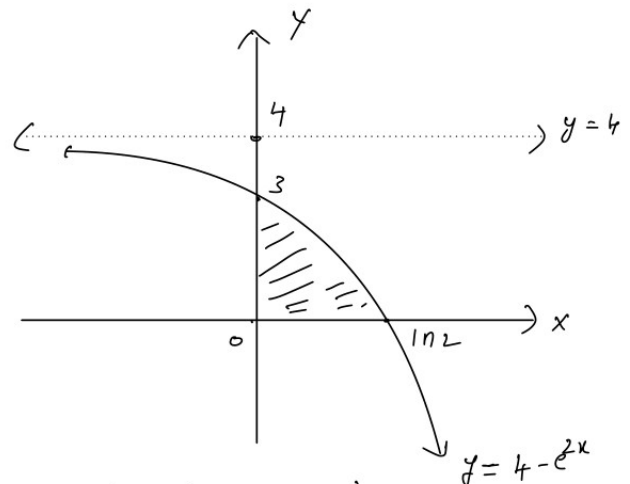
$$V_x = \frac{\pi}{4} (e^2 + 4 - e^{-2}) \text{ units}^3$$

Integration Involving Exponential Functions and Applications continued

16. (a) Find the x -intercept of the curve $y = 4 - e^{2x}$.
- (b) Hence find the area of the region bounded by the curve $y = 4 - e^{2x}$ and the coordinate axes.
- (c) Find the volume of the solid of revolution obtained when the region in part (b) is rotated about the x -axis.



(a) $x \rightarrow \text{int: } y = 0$
 $4 - e^{2x} = 0$
 $4 = e^{2x}$
 $\Rightarrow 2x = \ln 4$
 $x = \frac{\ln 4}{2}$
 $x = \ln 2$



(b) Area, $A = \int_0^{\ln 2} (4 - e^{2x}) dx$
 $= \left[4x - \frac{e^{2x}}{2} \right]_0^{\ln 2} = \left(4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right) - \left(4(0) - \frac{1}{2} e^{2(0)} \right)$
 $= 4 \ln 2 - \frac{1}{2} (4^2) + \frac{1}{2}$
 $A = \left(4 \ln 2 - \frac{3}{2} \right) \text{ units}^2$

(c) Volume, $V_x = \pi \int_0^{\ln 2} (4 - e^{2x})^2 dx = \pi \int_0^{\ln 2} (16 - 8e^{2x} + e^{4x}) dx$
 $V_x = \pi \left[16x - 8 \cdot \frac{e^{2x}}{2} + \frac{e^{4x}}{4} \right]_0^{\ln 2}$
 $= \pi \left[\left(16 \ln 2 - 4 e^{2 \ln 2} + \frac{1}{4} e^{4 \ln 2} \right) - \left(16 \cdot 0 - 4 e^{2 \cdot 0} + \frac{1}{4} e^{4 \cdot 0} \right) \right]$
 $= \pi \left[\left(16 \ln 2 - 16 + 4 \right) - \left(-4 + \frac{1}{4} \right) \right]$
 $= \pi \left[16 \ln 2 - 12 + \frac{15}{4} \right] = \pi \left[16 \ln 2 - 8 \frac{1}{4} \right] \text{ units}^3$